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Research on key technologies for fault knowledge acquisition of power communication equipment

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Abstract

With the development of Energy Internet and Smart Grid, power communication equipment has gradually become an important pillar supporting the information transmission. However, when equipment failure occurs, more and more attention has been paid to the rapid realization of equipment fault research and judgment from a large amount of equipment fault information. Based on the fault data of power communication equipment, a fault model of power communication equipment is designed in this paper. It combines BERT and BiLSTM-CRF model for knowledge extraction, and constructs the equipment fault knowledge graph to realize the visual display of power communication equipment. Through this model, fault knowledge extraction can be realized accurately, and a visual model of fault analysis can be formed, which can effectively improve the fault research speed and work efficiency of maintenance personnel.

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1. Introduction

Due to the continuous promotion and popularization of Energy Smart Grids, many new power information communication devices are continuously being developed and produced. These devices are used to transmit power grid data and sense power grid status. It has gradually become an indispensable part of power grid fault research and

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automatic operation and maintenance. However, there are many kinds of electric power communication equipment, and there are various fault types, phenomena and attributes. Therefore, how to construct the fault model of power communication equipment to achieve accurate equipment fault location and rapid query of equipment information is the basis to achieve equipment troubleshooting and ensure the normal operation of the power grid. In the field of artificial intelligence, due to the combination of natural language processing technology and deep learning, it is developing more and more rapidly. Through this technology, the structured and unstructured data of power communication equipment faults can be processed. According to the extracted knowledge, the knowledge graph is constructed to form the fault model of power communication equipment. By virtue of this model, power grid repair and inspection personnel can quickly locate the faults of power communication equipment and play an effective role in assisting decision-making.

Although the existing knowledge graph construction has begun to take shape, the extraction of entity relations is also very effective. However, it is less involved in power equipment, and the accuracy of entity relationship extraction still needs to be improved. Based on the fault processing text and fault log data of existing power communication equipment, a fault model of power communication equipment is designed in this paper. The fault text and data are pre-trained by introducing BERT structure. Then the joint entity-relation extraction from the corpus is realized with the help of BiLSTM-CRF model. Finally, the fault knowledge graph is constructed by using the graph database to visually describe the equipment fault handling mode, type and other attributes. It is helpful to help power grid fault repair personnel to assist decision-making and fault location quickly.

2. Related work

2.1. Knowledge graph

The concept of knowledge graph was put forward by Google as early as 2012 and applied to search engines to facilitate users to quickly get the information they need. Then encyclopedic atlas products such as Wikidata knowledge base and "YAGO" knowledge graph appeared abroad, and "knowledge cube" emerged in China, "CN-Dbpedia and Openkg.CN" of Fudan University and "SciKG, XKLORE" encyclopedia atlas of Tsinghua University. At the same time, the knowledge graph is also cross-studied with many disciplines, covering many fields, such as health care, education, transportation, e-commerce and so on. The key point and difficulty of constructing domain knowledge graph lies in dealing with the relationship between structured and unstructured, that is, the realization of named entity recognition, relationship extraction and graph construction.

At present, in the aspect of constructing knowledge graph, Lin [1] studies named entity recognition and entity relationship extraction of nephropathy electronic medical records, and proposes long-term and short-term memory network-conditional random field model to extract entities. Then use the convolution neural network model to extract entities and entity relations from the text and transfer them to the Neo4j database to build a knowledge graph. Qiao [2] proposed a knowledge mapping application framework for power grid fault handling decision-making, which condenses the unstructured content of power grid dispatching operations. According to business scenarios and grid fault handling requirements, classify domain knowledge. So as to realize the fault information analysis and discrimination.

2.2. Knowledge extraction

Knowledge extraction mainly consists of two parts, entity extraction and relationship extraction. In order to improve the accuracy of entity relationship extraction, Pierre [2] proposed a weak supervision method to guide the named entity recognition model without any tag data from the target domain. By training the hidden Markov model (HMM), considering the accuracy and confusion of each marking function, the noise marking function is unified into a single (probability) annotation. Finally, the cross-entropy loss is used to train the sequence labeling model on the unified tagging.

Tabassum [4] trains the BERT representation of computer programming corpus and puts forward a Soft-NER model, which effectively improves the F1 score of named entity recognition. Wei [5] proposed a cascaded binary labeling framework (CASREL). It uses a pre-trained BERT encoder to model the relationship as a function that

maps the subject in the sentence to the object, and deals with the problem of relation overlap. Kruiper [6] combines the output of narrow-IE system and Open-IE to realize semi-open relation extraction. First of all, the narrow-IE system is pre-trained with FOBIE, and then the relationship is extracted and its accuracy is compared. It can rely on the existing encyclopedia knowledge base and the third-party knowledge base to extract entity relations more accurately.

3. Technical architecture design of communication equipment knowledge acquisition

The construction of the fault model of power grid communication equipment is mainly designed from three parts. It mainly includes named entity recognition, entity relationship extraction and knowledge graph construction. Because the faults of power grid communication equipment are mostly text data, which belong to unstructured data, so the data foundation of the model is aimed at unstructured data. The following three parts of the implementation process are described. The architectural design is shown in Figure 1:

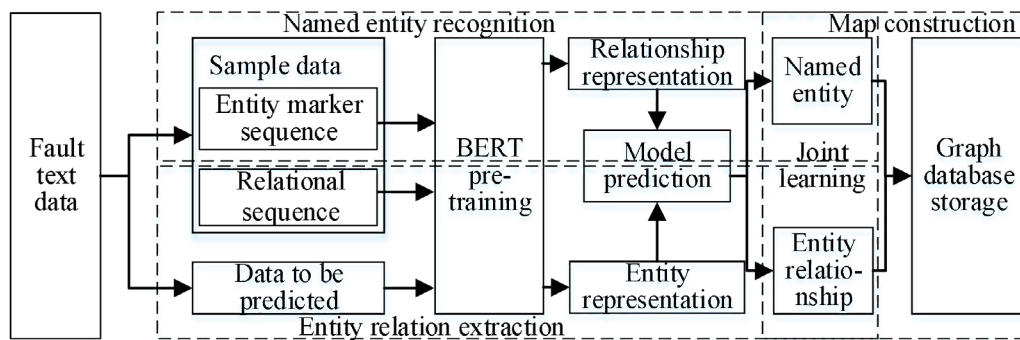


Fig. 1. Technical architecture design of communication equipment knowledge acquisition diagram

3.1. Named entity recognition

Named entity recognition is not only the key technology of constructing knowledge graph, but also the basis of fault model. In the named entity recognition part of the model, a pre-training model based on BERT is introduced to represent the word vector in low dimension. Then the BiLSTM neural network is used to mine the deep features of the word vector. Finally, CRF is used to obtain the sequence tagging and prediction of the corpus to achieve the accurate recognition of named entities.

3.2. Entity relation extraction

Entity relation extraction is based on named entity recognition and BERT pre-training model to encode the corpus. Then the BiLSTM layer parameters are shared by joint learning, and the entity tags predicted by CRF and the corpus of BiLSTM are used as the common input of relation extraction. Relationship extraction is also realized on the basis of named entity recognition.

3.3. Construction of knowledge graph

Through the named entity recognition and relational extraction of fault data, a large amount of entity-relational information obtained is stored in the form of graph database. Then collate, import and view the data, visually display the fault information, and realize the overall construction of the fault model of power information equipment.

4. Method for fault knowledge acquisition of communication equipment

According to the characteristics of fault text of power communication equipment, this paper adopts BIO standard strategy to label entity relations. The BiLSTM-CRF model based on BERT is introduced to extract knowledge from data text. Finally, the knowledge graph is constructed according to the extracted entity relationship. The construction process is shown in Figure 2. Next, each algorithm is described in detail.

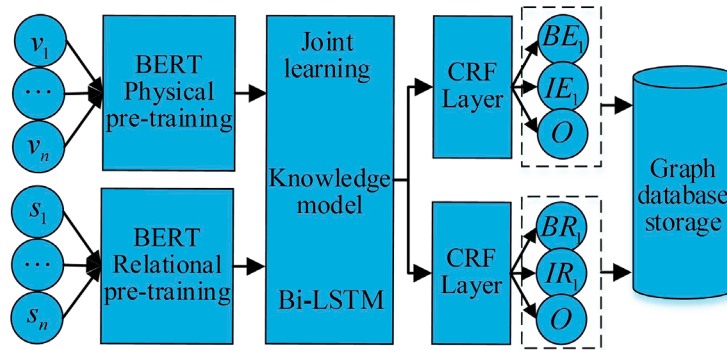


Fig. 2. Flow chart of fault model construction

4.1. BiLSTM-CRF knowledge extraction model based on BERT

- Pre-training Model of knowledge extraction based on BERT

Word embedding is the foundation of natural language processing technology, which can map a word to low-dimensional semantic space.

Through the encoder of bi-directional Transformer, the BERT model can obtain the information in both directions before and after the sentence at the same time. Transformer includes two parts: position coding and self-attention mechanism. The position coding can take into account the position vector related to t time and avoid the problem of word disorder. At the same time, the self-attention mechanism effectively solves the problem of long-term dependence of traditional RNN and CNN neural networks. When entering a sentence, both its encoder and decoder use Self-Attention to represent the text. The calculation process is shown in formula 1:

$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

K , V and Q are the output of the encoder and the input of attention in the decoder respectively. Because BERT and the training model have better word embedding effect and can effectively represent the text data, this model is used for pre-training for entity recognition and relationship extraction respectively.

- Joint Model based on BiLSTM Neural Network

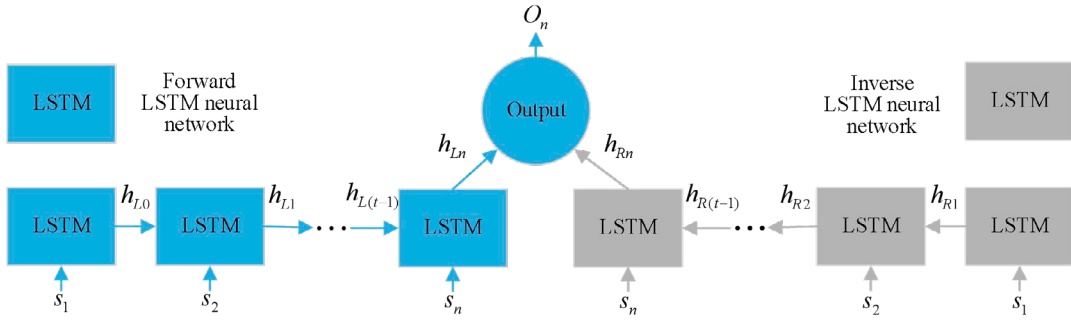


Fig. 3. Structure diagram of BiLSTM neural network

BiLSTM neural network, as shown in Figure 4, is composed of forward LSTM and backward LSTM. It processes sentences on the basis of BERT preprocessing model. This structure can better capture context information and effectively realize the prediction of entities and relationship tags. As shown in the figure, the neural network structure diagram is shown, in which $x(s_1, s_2, \dots, s_n)$ and $x'(s'_1, s'_2, \dots, s'_n)$ are inputs, which are obtained by double operation:

$$h_{Ln} = f(Ws_t + Vh_{L-1}) \quad (2)$$

$$h_{Rn} = f(Ws'_t + Vh_{R-1}) \quad (3)$$

Where f is the LSTM neuron activation function. h_{Ln} and h_{Rn} are the output information vectors after operation, respectively. W is the weight matrix from the hidden layer to the output layer. V is the hidden layer weight matrix calculated from the previous time to the next time. Finally, h_{Ln} and h_{Rn} are calculated through the activation function of the output layer.

$$O_n = g(W'h_{Ln} + W'h_{Rn}) \quad (4)$$

g is the activation function of the output layer and W is the weight matrix of the output layer.

- Sequence tagging Model based on CRF

Although BiLSTM neural network has been fully integrated with context information, the dependency information and label order between tags are not taken into account. Conditional random field model (Conditional Random Field) is proposed by Lafferty, which can add constraints to the predicted tags in the task of named entity and relationship extraction, so as to ensure the rationality of the prediction results. Such constraints are gradually learned as the data is input into the model training. The CRF layer calculates the score of the tag sequence through the formula:

$$score(x, y) = \sum_{i=1}^n P_i, y_i + \sum_{i=1}^{n+1} Q_{y_{i-1}, y_i}, \lim_{x \rightarrow \infty} y_i \quad (5)$$

Where A is the transformation matrix. P_i represents the score of the y_i th label corresponding to the character. Then use Softmax for normalized classification, such as:

$$P(y | x) = \frac{\exp(score(x, y))}{\sum y' \exp(score(x, y'))} \quad (6)$$

4.2. Fault knowledge storing based on Neo4j graph database

Through the knowledge extraction of the fault text, the corresponding entity and relation sequence are obtained. The next step is to use these data to build the fault model of power information and communication equipment to achieve a comprehensive description of the equipment fault forms and processing methods. This article uses the Neo4j diagram database to store entities and relationships. The steps for building a diagram database using Neo4j are shown in Figure 6:



Fig. 4. flowchart of fault knowledge graph construction

Step1: Converts the extracted entity and relational data storage form into csv format for import.

Step2: Uses Neo4j graph data language to store entity nodes first.

Step3: Imports relational data and establishes links between nodes.

Step4: Uses knowledge graph to display fault visualization model.

4.3. Knowledge-driven Fault diagnosis steps of communication equipment

In this paper, the standard strategy, BIO tagging method. The process of naming entity-relationship using BERT and training model is as follows:

Step1: Firstly, annotates the fault text data according to the annotation strategy according to the defined entity type and relation type. And in the labeling process to clean the data set, delete redundant useless characters, including newline characters, space characters and so on. Then the named entity labeling sequence and entity relationship labeling sequence are collected by sliding window segmentation, which are used as the input data for model training. Filter the data scale through the window and filter out the sentences with larger length.

Step2: Uses the marked and segmented corpus as the input of the pre-training model based on BERT to preprocess the corpus. Then input the processed named entity corpus into the BiLSTM-CRF model. The recognition accuracy of the model is continuously improved by adjusting the learning parameters and improving the number of training to optimize the named entity recognition model.

Step3: Uses joint learning to share parameters with the BiLSTM layers in the named entity recognition model. Input the entity relationship marker corpus processed by the BERT preprocessing model into the BiLSTM-CRF model, continuously optimize the entity relationship recognition models by adjusting the learning parameters and increasing the number of training, and improve the recognition accuracy of the model. Step4: cleans other sample data, and also uses sliding window to collect prediction data. Two trained BERT pre-trainings models and BiSTM-CRF model are used to predict the label of the corpus text. Finally, we get the corpus markers for all the fault texts.

Step4: According to the named entities and relationships marked by the model, this paper uses Neo4j graph database to construct knowledge graph. And then realize the visual display and convenient analysis of equipment faults.

5. Conclusion

In this paper, combined with the existing fault data text, warning log and other information of power communication equipment, a fault model of power communication equipment based on knowledge graph is designed. The contributions are as follows: (1) The model introduces the BERT pre-training model to preprocess the data. (2) Joint extraction of named entities and entity relationships based on the BiLSTM-CRF model. (3) Use Neo4j graph database to implement the visualization of power communication equipment faults. This model can help maintainers to quickly automated obstacles and improve the automatic operations.

Acknowledgement

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